CS 6515: Graduate Algorithms

Lecture 15: LP Relaxations and Deterministic Rounding

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Today's question. How can an optimal *fractional* solution help us find a good *integral* solution? We will first round Vertex Cover, then develop a 6-approximation for metric Facility Location.

1 The LP-rounding template

1. Formulate the discrete optimization problem as an integer program (IP).
2. Relax the integrality constraints to obtain a linear program (LP).
3. Solve the LP, obtaining an optimum x^* and a lower bound $^* \leq \mathrm{OPT}$ for minimization.
4. Round the fractional solution to a feasible integral one, bounding the cost increase.

The goal is to prove $\mathrm{cost}(\text{rounded solution}) \leq \alpha^* \leq \alpha\,\mathrm{OPT}$.

2 Vertex Cover: threshold rounding

Given $G = (V, E)$, introduce a variable x_v indicating whether vertex v is chosen. The integer program is

$$\begin{aligned} \text{minimize} \quad & \sum_{v \in V} x_v \\ \text{subject to} \quad & x_u + x_v \geq 1 \quad \forall \{u, v\} \in E, \\ & x_v \in \{0, 1\} \quad \forall v \in V. \end{aligned}$$

Relax to $0 \leq x_v \leq 1$. Let x^* be an optimal fractional solution, so $\sum_v x_v^* =^* \leq \mathrm{OPT}$.

Rounding rule. Return $C = \{v : x_v^* \geq 1/2\}$.

Theorem (Deterministic Rounding for Vertex Cover). Threshold rounding produces a vertex cover of size at most $2\,\mathrm{OPT}$.

Proof. For each edge $\{u, v\}$, the LP requires $x_u^* + x_v^* \geq 1$, so at least one endpoint has value at least $1/2$. Hence C covers every edge. Moreover, $\mathbf{1}[v \in C] \leq 2x_v^*$, and therefore

$$|C| = \sum_v \mathbf{1}[v \in C] \leq 2 \sum_v x_v^* = 2\,\mathrm{OPT}\,. \qquad \square$$

Connection to the matching algorithm. The LP dual (omitting redundant upper bounds) is

$$\begin{aligned} \text{maximize} \quad & \sum_{e \in E} y_e \\ \text{subject to} \quad & \sum_{e \ni v} y_e \leq 1 \quad \forall v \in V, \\ & y_e \geq 0 \quad \forall e \in E. \end{aligned}$$

A matching gives a feasible integral dual solution by setting $y_e = 1$ on matching edges and 0 elsewhere. Thus both lectures used lower-bound certificates, in two different forms.

Integrality gap. In the complete graph K_n, the integral optimum is $n-1$, while the fractional optimum is $n/2$ (set each $x_v = 1/2$; optimality also follows by summing the edge constraints). The ratio $2 - 2/n$ approaches 2. The LP itself cannot generally certify a better factor than 2.

3 Facility Location

Let F be candidate facilities and C clients. Opening facility $i \in F$ costs $f_i \geq 0$. Assigning client $j \in C$ to facility i costs $d_{ij} \geq 0$. Distances are induced by a metric on $F \cup C$, so the triangle inequality applies.

Choose facilities $S \subseteq F$ to open and connect each client to an open facility. Minimize

$$\sum_{i \in S} f_i + \sum_{j \in C} \min_{i \in S} d_{ij}.$$

Integer program. Let $x_i = 1$ mean facility i is open, and $y_{ij} = 1$ mean client j is assigned to i:

$$\begin{aligned} \text{minimize} \quad & \sum_{i \in F} f_i x_i + \sum_{i \in F, j \in C} d_{ij} y_{ij} \\ \text{subject to} \quad & \sum_{i \in F} y_{ij} = 1 \quad \forall j \in C, \\ & y_{ij} \leq x_i \quad \forall i \in F,\ j \in C, \\ & x_i, y_{ij} \in \{0, 1\}. \end{aligned}$$

Relax integrality to $0 \leq x_i, y_{ij} \leq 1$, and solve for (x^*, y^*). Define

$$F^* = \sum_i f_i x_i^*, \qquad D^* = \sum_{i,j} d_{ij} y_{ij}^*, \qquad OPT^* = F^* + D^* \leq \text{OPT}\,.$$

3.1 Deterministic Rounding for Facility Location

Let (x^*, y^*) be an optimal solution to the LP relaxation. Recall that x_i^* represents the fractional opening of facility i, and y_{ij}^* represents the fraction of client j assigned to facility i.

For each client $j \in C$, define

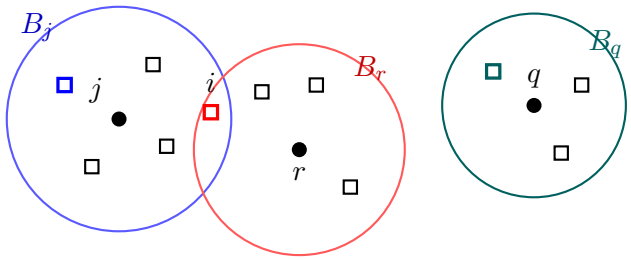
$$L_j = \sum_{i \in F} d_{ij} y_{ij}^*.$$

We interpret L_j as the average distance from client j to the facilities serving it in the fractional solution.

Define the ball

$$B_j = \{i \in F : d_{ij} \leq 2L_j\}.$$

Thus B_j contains all candidate facilities within distance $2L_j$ of client j.



The balls B_j and B_r overlap because they share a candidate facility i. The ball B_q is disjoint from both. Squares represent candidate facilities and dots represent clients.

3.2 The Rounding Algorithm

We process clients in nondecreasing order of L_j. The set R will contain the representatives of the balls selected by the algorithm.

Algorithm: Facility Location Rounding

1. Sort clients so that
$$L_{j_1} \leq L_{j_2} \leq \cdots \leq L_{j_{|C|}}.$$
2. Initialize $R = \emptyset$ and the set of open facilities $S = \emptyset$.
3. For each client j, in the above order:
   (a) If $B_j \cap B_r = \emptyset$ for every $r \in R$, choose the cheapest facility in B_j:
   $$i_j \in \arg\min_{i \in B_j} f_i.$$
   Open facility i_j, add i_j to S, add j to R, and assign j to i_j.
   (b) Otherwise, choose some $r \in R$ with $B_j \cap B_r \neq \emptyset$ and assign client j to the facility i_r opened for r.
4. Return the opened facilities and client assignments.

Notice that the balls corresponding to clients in R are pairwise disjoint. This property is essential for bounding the opening costs.

Next time we will prove that this algorithm yields a $6-$approximation.